

Group Expense Splitter

Phase 9 — Feature Completion Report

Module: Expense Tracking & Settlement Engine
Status: All Phase 9 requirements implemented and verified (QA passed)
Prepared for: GitHub Commit / Sprint Closure Documentation

This report documents the completion status of **Phase 9** of the Group Expense Splitter application. The module implements a full expense-tracking workflow covering **common (shared) expenses** and **custom expenses**, an **equal-split settlement engine**, per-member **balance summary computation**, and selective member participation via checkbox-based split groups. A live application screenshot is attached in this report as functional proof-of-work (Appendix A).

Category	Status	Notes
Common Expense Module	Complete	Auto-split across all members
Custom Expense Module	Complete	Selective member split via checkboxes
Balance Summary Engine	Complete	Owed / Owes computed per member
Paid-By Attribution	Complete	Displayed on every expense entry
Data Segregation	Complete	Common vs Custom kept in separate ledgers

1. Job Requirements — Completion Checklist

All assigned requirements for this module have been implemented, tested, and confirmed working end-to-end in the current build.

#	Requirement	Status
1	Add Common Expense (shared across all members)	Done
2	Balance Summary (auto-calculated, real-time)	Done
3	Add Custom Expense (independent expense entry)	Done
4	Select Specific Members (checkbox-based split group)	Done
5	Custom Expense List (chronological log)	Done
6	Custom Balance Summary (isolated ledger)	Done
7	"Paid By" Display (attribution per transaction)	Done
8	Common vs Custom Expenses kept separate (data model)	Done
9	Equal Split Calculation (division logic / rounding)	Done

2. Technical Notes

- **Split logic:** equal-division algorithm — total amount is divided by the count of selected members (checkbox-driven subset), not the full group, enabling per-expense custom participation.
- **Balance computation:** a running ledger (owed vs. owes) is derived per user by aggregating paid-amount against fair-share across all transactions in that ledger.
- **State segregation:** Common and Custom expenses are maintained as two independent data arrays/collections, each with its own balance-summary reducer — preventing cross-contamination of totals.
- **Attribution:** every expense record stores a *paidBy* field, rendered inline in both the Expenses list and Custom Expenses list.
- **UI:** checkbox group under "Split Between" drives the participant array passed into the split function at submit time.

3. Phase 9 Checklist

Item	Status
Common expense	Done
Custom expense	Done
Expense history	Done
Paid by	Done
Split between selected members	Done
Balance summary	Done
Custom balance summary	Done

All Phase 9 milestones are marked complete. The module is stable and ready for commit / merge into main.

4. Optional Improvements — Future Roadmap

The following items are **not required for Phase 9** but are logged as backlog items for future iterations:

- **Edit Expense** — inline update of an existing expense entry (amount, description, participants).
- **Delete Expense** — soft/hard delete with balance-summary recalculation on removal.
- **Expense Categories Filter** — tagging (food, travel, stay, misc.) with filter/search UI.
- **Settlement Suggestions** — a debt-simplification algorithm to generate minimal transactions, e.g. "A pays B ■200".
- **Export Expenses** — PDF/CSV export of expense history and balance summaries for offline record-keeping.

Appendix A — Functional Proof of Work

Live application screenshot showing the implemented features: Expenses list with paid-by attribution, Balance Summary, Add Custom Expense form with member-selection checkboxes, Custom Expenses list, and Custom Balance Summary — all rendering correctly with live computed values.

The screenshot displays the Expense Splitter app interface, which is organized into several sections:

- Expenses:** A list showing "hotel-day 1 — paid by priyanshu" with a total value of ₹1500.
- Balance summary:** A table showing the balance for three users: priyanshu (is owed ₹1125), PRIYANSHU KUMAR (owes ₹375), test user (owes ₹375), and sk (owes ₹375).
- Add Custom Expense:** A form with input fields for "Description" and "Amount". Below these is a "Split Between" section with checkboxes for PRIYANSHU KUMAR, test user, priyanshu, and sk. A green "Add Custom Expense" button is at the bottom.
- Custom Expenses:** A list showing "coffee — paid by priyanshu" (₹40) and "tea — paid by priyanshu" (₹20).
- Custom Balance Summary:** A table showing the custom balance for three users: priyanshu (is owed ₹50), sk (owes ₹10), and test user (owes ₹40).

User	Status	Amount
priyanshu	is owed	₹1125
PRIYANSHU KUMAR	owes	₹375
test user	owes	₹375
sk	owes	₹375

User	Status	Amount
priyanshu	is owed	₹50
sk	owes	₹10
test user	owes	₹40

Fig. 1 — Expense Splitter app UI (common + custom expense modules, live data)